SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Cover

	TEST REPORT DOCUMENT				
Project Name	VISTA: Vision-Language-Action System for Tabletop Manipulation with Dual Robot Arms			Creator	
Project Code	FA26SE183			Issue Date	27/10/2026
Document Code	SEP490_FA26SE183_R2_TestReport_v0.2			Version	
Record of change					
Effective Date	Version	Change Item	*A,D,M	Change description	Reference
	27/10/2026	0.2	M	Reconciled proposed test inventory; no execution result claimed	BM01; traceability matrix; state-transition inventory

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Test Cases

			TEST CASE LIST	
	Project Name		VISTA	
	Project Code		FA26SE183	
	Test Environment Setup Description		Isolated role-seeded NestJS API + controlled Supabase/PostgreSQL; CI/staging not yet verified	
	No	Function Name	Sheet Name	Description
	1	TC-FR01-01 — FR-01 Role-governed access	Feature 1	Happy path: access a role-scoped API
	2	TC-FR01-02 — FR-01 Role-governed access	Feature 1	Validation boundary: access a role-scoped API
	3	TC-FR01-03 — FR-01 Role-governed access	Feature 1	Authorization boundary: cross-role/cross-organization access
	4	TC-FR02-01 — FR-02 Classroom robot allocation	Feature 2	Happy path: allocate an entitled robot to an owned classroom
	5	TC-FR02-02 — FR-02 Classroom robot allocation	Feature 2	Validation boundary: allocate an entitled robot to an owned classroom
	6	TC-FR02-03 — FR-02 Classroom robot allocation	Feature 2	Authorization boundary: allocate an unentitled or foreign robot
	7	TC-FR03-01 — FR-03 Assignment publication	Feature 3	Happy path: create and publish classroom work
	8	TC-FR03-02 — FR-03 Assignment publication	Feature 3	Validation boundary: create and publish classroom work
	9	TC-FR03-03 — FR-03 Assignment publication	Feature 3	Authorization boundary: publish into an unowned classroom
	10	TC-FR04-01 — FR-04 Immutable program submission	Feature 4	Happy path: submit an owned immutable program version
	11	TC-FR04-02 — FR-04 Immutable program submission	Feature 4	Validation boundary: submit an owned immutable program version
	12	TC-FR04-03 — FR-04 Immutable program submission	Feature 4	Authorization boundary: submit another learner's version
	13	TC-FR05-01 — FR-05 Schematic experiment	Feature 5	Happy path: run and persist a schematic experiment
	14	TC-FR05-02 — FR-05 Schematic experiment	Feature 5	Validation boundary: run and persist a schematic experiment
	15	TC-FR05-03 — FR-05 Schematic experiment	Feature 5	Authorization boundary: run outside the learner scope
	16	TC-FR06-01 — FR-06 Physical execution boundary	Feature 6	Happy path: request an eligible bounded physical run

			TEST CASE LIST	
	17	TC-FR06-02 — FR-06 Physical execution boundary	Feature 6	Validation boundary: request an eligible bounded physical run
	18	TC-FR06-03 — FR-06 Physical execution boundary	Feature 6	Authorization boundary: report success after unavailable/unsafe execution
	19	TC-FR07-01 — FR-07 Assessment and feedback	Feature 7	Happy path: assess work and publish/read scoped feedback
	20	TC-FR07-02 — FR-07 Assessment and feedback	Feature 7	Validation boundary: assess work and publish/read scoped feedback
	21	TC-FR07-03 — FR-07 Assessment and feedback	Feature 7	Authorization boundary: review or read a foreign submission
	22	TC-FR08-01 — FR-08 AI/model provenance	Feature 8	Happy path: read traceable AI/model/dataset metadata
	23	TC-FR08-02 — FR-08 AI/model provenance	Feature 8	Validation boundary: read traceable AI/model/dataset metadata
	24	TC-FR08-03 — FR-08 AI/model provenance	Feature 8	Authorization boundary: report unavailable provider work as successful
	25	TC-WF01-01 — Workflow Scenarios	Workflow Scenarios	WF-01 school administration and robot allocation
	26	TC-WF02-01 — Workflow Scenarios	Workflow Scenarios	WF-02 learning assignment and submission
	27	TC-WF03-01 — Workflow Scenarios	Workflow Scenarios	WF-03 experiment execution and evaluation
	28	TC-WF04-01 — Workflow Scenarios	Workflow Scenarios	WF-04 teacher review and AI/data evidence
	29	ST-EXP-01 — State Transitions	State Transitions	Requested → Authorized
	30	ST-EXP-02 — State Transitions	State Transitions	Requested → Rejected
	31	ST-EXP-03 — State Transitions	State Transitions	Authorized → Queued
	32	ST-EXP-04 — State Transitions	State Transitions	Queued → Running
	33	ST-EXP-05 — State Transitions	State Transitions	Queued → Cancelled
	34	ST-EXP-06 — State Transitions	State Transitions	Running → Completed
	35	ST-EXP-07 — State Transitions	State Transitions	Running → Failed
	36	ST-EXP-08 — State Transitions	State Transitions	Running → Cancelled
	37	ST-EXP-09 — State Transitions	State Transitions	Completed → Archived
	38	ST-EXP-10 — State Transitions	State Transitions	Failed → Archived
	39	ST-EXP-11 — State Transitions	State Transitions	Cancelled → Archived
	40	ST-EXP-12 — State Transitions	State Transitions	Requested → Completed
	41	ST-EXP-13 — State Transitions	State Transitions	Archived → Running
	42	ST-RUN-01 — State Transitions	State Transitions	Idle → Ready Check

			TEST CASE LIST	
	43	ST-RUN-02 — State Transitions	State Transitions	Ready Check → Reserved
	44	ST-RUN-03 — State Transitions	State Transitions	Ready Check → Not Ready
	45	ST-RUN-04 — State Transitions	State Transitions	Reserved → Executing
	46	ST-RUN-05 — State Transitions	State Transitions	Executing → Uploading Evidence
	47	ST-RUN-06 — State Transitions	State Transitions	Uploading Evidence → Completed
	48	ST-RUN-07 — State Transitions	State Transitions	Executing → Stopped
	49	ST-RUN-08 — State Transitions	State Transitions	Executing → Failed
	50	ST-RUN-09 — State Transitions	State Transitions	Not Ready → Idle
	51	ST-RUN-10 — State Transitions	State Transitions	Stopped → Idle
	52	ST-RUN-11 — State Transitions	State Transitions	Failed → Idle
	53	ST-RUN-12 — State Transitions	State Transitions	Completed → Idle
	54	ST-RUN-13 — State Transitions	State Transitions	Idle → Executing
	55	ST-RUN-14 — State Transitions	State Transitions	Completed → Executing
	56	ST-SUB-01 — State Transitions	State Transitions	Draft → In Progress
	57	ST-SUB-02 — State Transitions	State Transitions	In Progress → Submitted
	58	ST-SUB-03 — State Transitions	State Transitions	Submitted → Under Review
	59	ST-SUB-04 — State Transitions	State Transitions	Under Review → Accepted
	60	ST-SUB-05 — State Transitions	State Transitions	Under Review → Returned
	61	ST-SUB-06 — State Transitions	State Transitions	Under Review → Rejected
	62	ST-SUB-07 — State Transitions	State Transitions	Accepted → Published Feedback
	63	ST-SUB-08 — State Transitions	State Transitions	Returned → In Progress
	64	ST-SUB-09 — State Transitions	State Transitions	Rejected → Published Feedback
	65	ST-SUB-10 — State Transitions	State Transitions	Draft → Accepted
	66	ST-SUB-11 — State Transitions	State Transitions	Published Feedback → In Progress
	67	ST-ASG-01 — State Transitions	State Transitions	Draft → Scheduled
	68	ST-ASG-02 — State Transitions	State Transitions	Scheduled → Published
	69	ST-ASG-03 — State Transitions	State Transitions	Published → Open
	70	ST-ASG-04 — State Transitions	State Transitions	Open → Closed
	71	ST-ASG-05 — State Transitions	State Transitions	Closed → Archived
	72	ST-ASG-06 — State Transitions	State Transitions	Scheduled → Cancelled
	73	ST-ASG-07 — State Transitions	State Transitions	Published → Cancelled
	74	ST-ASG-08 — State Transitions	State Transitions	Cancelled → Archived
	75	ST-ASG-09 — State Transitions	State Transitions	Draft → Open

			TEST CASE LIST	
	76	ST-ASG-10 — State Transitions	State Transitions	Archived → Published
	77	TC-NFR01-01 — NFR	NFR	Authorization isolation
	78	TC-NFR02-01 — NFR	NFR	Runtime safety/reliability
	79	TC-NFR03-01 — NFR	NFR	API responsiveness
	80	TC-NFR04-01 — NFR	NFR	Runtime acknowledgement
	81	TC-NFR05-01 — NFR	NFR	Execution traceability
	82	TC-NFR06-01 — NFR	NFR	Test automation

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Test Statistics

	TEST STATISTICS					
	Project Name	VISTA		Creator		
	Project Code	FA26SE183		Reviewer/Approver		
	Document Code			Issue Date		
	Notes	ex: Release 1 includes 2 modules: Module1 and Module2				
	No	Module code	Passed	Failed	Pending	N/A
	1	Feature 1				
	2	Feature 2				
	3	Feature 3				
	4	Feature 4				
	5	Feature 5				
	6	Feature 6				
	7	Feature 7				
	8	Feature 8				
	9	Workflow Scenarios				
	10	State Transitions				
	11	NFR				
		Sub total				
		Planned inventory			cases (expected 82)	
		Execution status		PENDING — not evidence of execution		

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 1

Feature	FR-01 Role-governed access													
Test requirement	access a role-scoped API													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR01-01	Happy path: access a role-scoped API	Authenticate as all roles; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR01-02	Validation boundary: access a role-scoped API	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-01 Role-governed access													
TC-FR01-03	Authorization boundary: cross-role/cross-organization access	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 2

Feature	FR-02 Classroom robot allocation													
Test requirement	allocate an entitled robot to an owned classroom													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function D														
TC-FR02-01	Happy path: allocate an entitled robot to an owned classroom	Authenticate as school admin; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR02-02	Validation boundary: allocate an entitled robot to an owned classroom	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-02 Classroom robot allocation													
TC-FR02-03	Authorization boundary: allocate an unentitled or foreign robot	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 3

Feature	FR-03 Assignment publication													
Test requirement	create and publish classroom work													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR03-01	Happy path: create and publish classroom work	Authenticate as teacher; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR03-02	Validation boundary: create and publish classroom work	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-03 Assignment publication													
TC-FR03-03	Authorization boundary: publish into an unowned classroom	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 4

Feature	FR-04 Immutable program submission													
Test requirement	submit an owned immutable program version													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR04-01	Happy path: submit an owned immutable program version	Authenticate as student; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR04-02	Validation boundary: submit an owned immutable program version	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-04 Immutable program submission													
TC-FR04-03	Authorization boundary: submit another learner's version	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 5

Feature	FR-05 Schematic experiment													
Test requirement	run and persist a schematic experiment													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR05-01	Happy path: run and persist a schematic experiment	Authenticate as student; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR05-02	Validation boundary: run and persist a schematic experiment	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-05 Schematic experiment													
TC-FR05-03	Authorization boundary: run outside the learner scope	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 6

Feature	FR-06 Physical execution boundary													
Test requirement	request an eligible bounded physical run													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR06-01	Happy path: request an eligible bounded physical run	Authenticate as student/runtime; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR06-02	Validation boundary: request an eligible bounded physical run	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-06 Physical execution boundary													
TC-FR06-03	Authorization boundary: report success after unavailable/unsafe execution	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 7

Feature	FR-07 Assessment and feedback													
Test requirement	assess work and publish/read scoped feedback													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR07-01	Happy path: assess work and publish/read scoped feedback	Authenticate as teacher/student; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR07-02	Validation boundary: assess work and publish/read scoped feedback	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-07 Assessment and feedback													
TC-FR07-03	Authorization boundary: review or read a foreign submission	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Feature 8

Feature	FR-08 AI/model provenance													
Test requirement	read traceable AI/model/dataset metadata													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-FR08-01	Happy path: read traceable AI/model/dataset metadata	Authenticate as student/teacher; prepare valid scoped data; perform the operation.	Request succeeds and the intended record/state is persisted.	Seeded authorized identity and valid domain data	Pending			Pending			Pending			Proposed; not executed
TC-FR08-02	Validation boundary: read traceable AI/model/dataset metadata	Submit a required field missing or an invalid state/value.	Request is rejected with a deterministic validation/domain error; no partial write.	Seeded authorized identity	Pending			Pending			Pending			Proposed; not executed

Feature	FR-08 AI/model provenance													
TC-FR08-03	Authorization boundary: report unavailable provider work as successful	Use a different role, learner, classroom, or organization against the same operation.	Request is denied without leaking or changing out-of-scope data.	Seeded second organization and unauthorized identity	Pending			Pending			Pending			Proposed; not executed

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: Workflow Scenarios

Feature	Workflow Scenarios													
Test requirement	Four mandatory main workflow scenarios													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-WF01-01	WF-01 school administration and robot allocation	Open owned classroom; select eligible robot; replace allocation.	Exactly one visible scoped allocation is persisted.	Seeded school admin, classroom, entitlements	Pending			Pending			Pending			Main workflow
TC-WF02-01	WF-02 learning assignment and submission	Teacher publishes work; learner starts attempt and submits owned program version.	Published assignment and immutable submission are traceable.	Seeded teacher, learner, classroom	Pending			Pending			Pending			Main workflow
TC-WF03-01	WF-03 experiment execution and evaluation	Create schematic experiment; run; evaluate; retrieve result/evidence.	Truthful terminal state, result, and evidence are persisted.	Seeded eligible attempt and experiment	Pending			Pending			Pending			Main workflow; physical E2E remains blocked

Feature	Workflow Scenarios													
TC-WF04-01	WF-04 teacher review and AI/data evidence	Teacher assesses; publishes feedback; learner reads and reflects.	Assessment, feedback and reflection remain scoped and traceable.	Seeded submitted work and authorized teacher	Pending			Pending			Pending			Main workflow

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: State Transitions

Feature	State Transitions													
Test requirement	40 valid and 8 invalid state-transition tests													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
ST-EXP-01	Requested → Authorized	Attempt transition Requested → Authorized.	Allowed when scope and permission pass	Entity is in Requested state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-02	Requested → Rejected	Attempt transition Requested → Rejected.	Allowed when scope or permission fails	Entity is in Requested state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-03	Authorized → Queued	Attempt transition Authorized → Queued.	Allowed after resource allocation	Entity is in Authorized state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-04	Queued → Running	Attempt transition Queued → Running.	Allowed when runtime is available	Entity is in Queued state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition

Feature	State Transitions													
ST-EXP-05	Queued → Cancelled	Attempt transition Queued → Cancelled.	Allowed before run	Entity is in Queued state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-06	Running → Completed	Attempt transition Running → Completed.	Allowed on successful terminal result	Entity is in Running state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-07	Running → Failed	Attempt transition Running → Failed.	Allowed on runtime error	Entity is in Running state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-08	Running → Cancelled	Attempt transition Running → Cancelled.	Allowed on stop or timeout	Entity is in Running state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-09	Completed → Archived	Attempt transition Completed → Archived.	Allowed by retention/close action	Entity is in Completed state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-10	Failed → Archived	Attempt transition Failed → Archived.	Allowed after outcome recording	Entity is in Failed state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-11	Cancelled → Archived	Attempt transition Cancelled → Archived.	Allowed after outcome recording	Entity is in Cancelled state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-EXP-12	Requested → Completed	Attempt transition Requested → Completed.	Rejected because authorization/execution were skipped	Entity is in Requested state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required

Feature	State Transitions													
ST-EXP-13	Archived → Running	Attempt transition Archived → Running.	Rejected because archived is terminal	Entity is in Archived state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required
ST-RUN-01	Idle → Ready Check	Attempt transition Idle → Ready Check.	Allowed on run command	Entity is in Idle state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-02	Ready Check → Reserved	Attempt transition Ready Check → Reserved.	Allowed when safety checks pass	Entity is in Ready Check state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-03	Ready Check → Not Ready	Attempt transition Ready Check → Not Ready.	Allowed when safety checks fail	Entity is in Ready Check state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-04	Reserved → Executing	Attempt transition Reserved → Executing.	Allowed when execution begins	Entity is in Reserved state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-05	Executing → Uploading Evidence	Attempt transition Executing → Uploading Evidence.	Allowed after task completion	Entity is in Executing state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-06	Uploading Evidence → Completed	Attempt transition Uploading Evidence → Completed.	Allowed after artifact persistence	Entity is in Uploading Evidence state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition

Feature	State Transitions													
ST-RUN-07	Executing → Stopped	Attempt transition Executing → Stopped.	Allowedon emergency stop	Entity is in Executing state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-08	Executing → Failed	Attempt transition Executing → Failed.	Allowed on execution error	Entity is in Executing state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-09	Not Ready → Idle	Attempt transition Not Ready → Idle.	Allowed after reset/retry	Entity is in Not Ready state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-10	Stopped → Idle	Attempt transition Stopped → Idle.	Allowed after resource release	Entity is in Stopped state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-11	Failed → Idle	Attempt transition Failed → Idle.	Allowed after recovery/release	Entity is in Failed state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-12	Completed → Idle	Attempt transition Completed → Idle.	Allowed when session closes	Entity is in Completed state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-RUN-13	Idle → Executing	Attempt transition Idle → Executing.	Rejected because readiness/reservation were skipped	Entity is in Idle state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required
ST-RUN-14	Completed → Executing	Attempt transition Completed → Executing.	Rejected because a new authorized session is required	Entity is in Completed state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required

Feature	State Transitions													
ST-SUB-01	Draft → In Progress	Attempt transition Draft → In Progress.	Allowed when learner starts work	Entity is in Draft state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-02	In Progress → Submitted	Attempt transition In Progress → Submitted.	Allowed on eligible submission	Entity is in In Progress state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-03	Submitted → Under Review	Attempt transition Submitted → Under Review.	Allowed when authorized teacher opens review	Entity is in Submitted state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-04	Under Review → Accepted	Attempt transition Under Review → Accepted.	Allowed when criteria are met	Entity is in Under Review state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-05	Under Review → Returned	Attempt transition Under Review → Returned.	Allowed when revision is requested	Entity is in Under Review state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-06	Under Review → Rejected	Attempt transition Under Review → Rejected.	Allowed for invalid/late submission under agreed rule	Entity is in Under Review state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-07	Accepted → Published Feedback	Attempt transition Accepted → Published Feedback.	Allowed when assessment is released	Entity is in Accepted state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition

Feature	State Transitions													
ST-SUB-08	Returned → In Progress	Attempt transition Returned → In Progress.	Allowed for resubmission	Entity is in Returned state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-09	Rejected → Published Feedback	Attempt transition Rejected → Published Feedback.	Allowed when rejection feedback is released	Entity is in Rejected state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-SUB-10	Draft → Accepted	Attempt transition Draft → Accepted.	Rejected because submission/review were skipped	Entity is in Draft state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required
ST-SUB-11	Published Feedback → In Progress	Attempt transition Published Feedback → In Progress.	Rejected unless a new attempt is created	Entity is in Published Feedback state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required
ST-ASG-01	Draft → Scheduled	Attempt transition Draft → Scheduled.	Allowed when teacher schedules work	Entity is in Draft state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-02	Scheduled → Published	Attempt transition Scheduled → Published.	Allowed on publish action	Entity is in Scheduled state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-03	Published → Open	Attempt transition Published → Open.	Allowed when availability begins	Entity is in Published state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-04	Open → Closed	Attempt transition Open → Closed.	Allowed when deadline/close time is reached	Entity is in Open state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition

Feature	State Transitions													
ST-ASG-05	Closed → Archived	Attempt transition Closed → Archived.	Allowed by archival action	Entity is in Closed state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-06	Scheduled → Cancelled	Attempt transition Scheduled → Cancelled.	Allowed before publication	Entity is in Scheduled state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-07	Published → Cancelled	Attempt transition Published → Cancelled.	Allowed on withdrawal under agreed rule	Entity is in Published state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-08	Cancelled → Archived	Attempt transition Cancelled → Archived.	Allowed after record retention	Entity is in Cancelled state with a seeded authorized actor.	Pending			Pending			Pending			Valid transition
ST-ASG-09	Draft → Open	Attempt transition Draft → Open.	Rejected because scheduling/publication were skipped	Entity is in Draft state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required
ST-ASG-10	Archived → Published	Attempt transition Archived → Published.	Rejected because archived is terminal	Entity is in Archived state with a seeded authorized actor.	Pending			Pending			Pending			Invalid transition; rejection required

SEP490_FA26SE183_R2_TestReport.xlsx — Sheet: NFR

Feature	NFR													
Test requirement	Six measurable non-functional requirements													
Number of TCs														
Testing Round	Passed	Failed	Pending	N/A										
Round 1														
Round 2														
Round 3														
Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Round 3	Test date	Tester	Note
Function A														
TC-NFR01-01	Authorization isolation	Run the predefined cross-role and cross-organization negative suite.	100% of forbidden accesses are rejected without disclosure.	Two seeded organizations and all roles	Pending			Pending			Pending			Target; execution pending
TC-NFR02-01	Runtime safety/reliability	Exercise duplicate, unavailable, timeout, stop and communication-loss cases.	No false successful terminal result; error/terminal evidence exists.	Mock runtime; hardware test separately supervised	Pending			Pending			Pending			Mock portion proposed; hardware blocked
TC-NFR03-01	API responsiveness	Run a time stamped authenticated API load test on declared staging data.	Ordinary API p95 is at most 3 seconds.	Declared staging configuration and dataset	Pending			Pending			Pending			Target; execution pending

Feature	NFR													
TC-NFR04-01	Runtime acknowledgement	Measure accepted physical command to acknowledgement or explicit timeout.	Acknowledgement or timeout occurs within 5 seconds.	Instrumented client/runtime	Pending			Pending			Pending			Target; physical environment blocked
TC-NFR05-01	Execution traceability	Query executed runs for actor, activity, resource/configuration, timestamps and terminal state.	100% of sampled executed runs contain every required field.	Seeded executions	Pending			Pending			Pending			Target; execution pending
TC-NFR06-01	Test automation	Reconcile committed inventory with tagged CI logs.	At least 50% automated; main workflows 100%; total pass rate at least 90%.	Valid review-2 tag and CI logs	Pending			Pending			Pending			Target; blocked until tagged revision